

Learning Optimal Strategies in Tic-Tac-Toe using Q-Learning

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1 Introduction

Reinforcement Learning (RL) has demonstrated strong capability in learning optimal strategies through interaction with an environment. In this project, we focus on applying Q-Learning, a classical model-free RL algorithm, to the game of Tic-Tac-Toe. While the game itself is simple and fully observable, it provides a structured environment to study convergence behavior, exploration strategies, and policy optimality.

Unlike purely implementation-based approaches, this project aims to investigate whether Q-Learning can consistently learn an optimal (unbeatable) strategy and how different design choices affect its performance.

2 Problem Formulation

We formulate Tic-Tac-Toe as a Markov Decision Process (MDP), defined by (S, A, P, R, γ) :

- S : all possible board configurations,
- A : available positions for placing a mark,
- P : deterministic state transitions given an action,
- R : reward function based on game outcomes,
- γ : discount factor.

The agent plays against an opponent (random or learning-based), and the objective is to learn a policy π^* that maximizes expected cumulative reward: $\mathbb{E} \left[\sum_{t=0}^T \gamma^t R(s_t, a_t) \right]$.

3 Related Work

Wang et al. (2019) assessed table-based Q-learning across small-board games and proposed a dynamic ϵ strategy, as well as QM-learning, to improve convergence [1]. Upadhyay (2021) analyzed the impact of different teaching opponents and showed that self-play Q-learning can converge without requiring an expert teacher [2].

How our project differs: While prior work focuses on cross-game generality or algorithmic enhancements, our project isolates the effects of training paradigms (random opponent vs. self-play) and exploration decay schedules (fixed, linear, exponential) on convergence and policy optimality within Tic-Tac-Toe. This focused scope enables more granular ablations and hyperparameter analysis.

4 Methodology

We implement the Q-Learning algorithm to estimate the optimal action-value function $Q(s, a)$: $Q(s, a) \leftarrow Q(s, a) + \alpha [R(s, a) + \gamma \max_{a'} Q(s', a') - Q(s, a)]$.

An ϵ -greedy policy is used to balance exploration and exploitation. We will encode board states efficiently and maintain a Q-table mapping state-action pairs to values.

To enhance the analysis, we will consider multiple training settings:

- Training against a random opponent,
- Self-play (Q-learning agent vs Q-learning agent),
- Different exploration strategies (fixed ϵ , linearly decaying ϵ , exponentially decaying ϵ).

5 Evaluation Plan

5.1 Evaluation Metrics

- **Cumulative Reward:** accumulated reward per episode, tracking learning progress.
- **Convergence Speed:** episodes needed for performance to stabilize.
- **Policy Optimality:** accuracy against a Minimax-optimal baseline (percentage of optimal moves).
- **Exploration Efficiency:** proportion of random actions taken during training under each ϵ schedule.

5.2 Experimental Settings

- **Opponents:** random agent (uniform legal moves) and self-play (agent vs. copy).
- **Hyperparameters:** ϵ , learning rate α , discount factor γ , and number of episodes will be varied for robustness checks.
- **Evaluation protocol:** every K training episodes, the current policy is evaluated for N_{eval} games against a fixed baseline (random or Minimax) with $\epsilon = 0$.

5.3 Expected Results

We expect that Q-Learning will learn a near-optimal strategy under appropriate settings. Self-play is anticipated to yield more robust policies than training against a random opponent, and decaying exploration strategies should accelerate convergence compared to a fixed ϵ .

6 Conclusion

This project aims to bridge implementation and analysis by applying Q-Learning to a classical game and systematically studying the convergence behavior, policy optimality, and the impact of training paradigms and exploration strategies. The findings will contribute to a deeper understanding of reinforcement learning dynamics in structured deterministic environments.

References

- [1] H. Wang, M. Emmerich, and A. Plaat. Assessing the Potential of Classical Q-learning in General Game Playing. *arXiv:1810.06078*, 2019.
- [2] I. Upadhyay. Analysis of Q-Learning Based Game Playing Agents for Abstract Board Games with Increasing State-Space Complexity. Master’s thesis, Miami University, 2021.